2023 WSMO Accuracy Round

SMO Team

AR 1: Let $x = \sqrt{69 + \sqrt{69 + \sqrt{69 \ldots}}}$. Find the value of $(2x - 1)^2$.

AR 2: When Bob is in precalculus, he gets bored and writes all the permutations in "precal". Since he is not very smart, it takes him 5 seconds to write each permutation. When Bob advances to calculus, he gets bored and writes all the permutations in "calculus". He is smart and can now write each permutation in 2 seconds. Find the positive difference in minutes between the time it takes for him to write the permutations of "precal" and "calculus".

AR 3: $f(x) = x^3 - 8x^2 + 10x - 4$ has complex roots a, b, and c. Denote $P(n) = a^n + b^n + c^n$. Find $P(-1)P(0)P(1)$.

AR 4: Ten people, including Bob and his 3 friends, stand in a line in random order. Given that Bob is not at either end, the probability that the people directly in front of and behind Bob are both his friends can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

AR 5: Bob flips $x+1$ coins and Bobby flips x coins, where x is an integer chosen uniformly at random from $[27, 100]$. The expected probability that Bob gets more heads than Bobby can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

AR 6: In quadrilateral $ABCD$, there exists a point O such that $AO = BO = CO = DO$ and $\angle(AOB) + \angle(COD) = 120^\circ$. Let K, L, M, N be the feet of the perpendiculars from A to BD, B to AC, C to BD, and D to AC. If $[ABCD] = 20$, find $([KLMN])^2$.

AR 7: How many ordered triplets of integers (a, b, c) satisfy $a^2 + 2ab + b^2 = c^2 - 6c + 9$ and $-2 \leq a, b, c \leq 7$?

AR 8: Let $f(x) = x^3 - 3x^2 + 4x + 5$ have complex roots a, b, and c. Then,

$$\frac{1}{a^2 + b^2} + \frac{1}{a^2 + c^2} + \frac{1}{b^2 + c^2}$$

can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

AR 9: Circles ω_1 and ω_2 with radii 1 and 4, respectively, are externally tangent to each other. The diameters of ω_1 and ω_2 are AB and CD, respectively, such that $AB \parallel CD$ and BD is a common external tangent to the circles. A third circle ω_3 passes through A and C and is tangent to BD. The minimum possible value of the radius of ω_3 can be expressed as $\frac{a - b\sqrt{c}}{d}$, where a, b, c, and d are positive integers, c is not divisible by the square of any prime, and $\gcd(a, b, d) = 1$. Find $a + b + c + d$.

AR 10: In regular tetrahedron T of side length 12, let S_1 be the sphere inscribed in T and let S_2 be the sphere circumscribed around T. Let R be a rectangular prism such that all points on S_1 lie strictly inside or are touching R and all points on R lie strictly inside or are touching S_2. The minimum possible volume of R can be expressed as $m\sqrt{n}$, where m and n are positive integers and n is not divisible by the square of any prime. Find $m+n$.